

DATA 615 - Regression: Homework - 3

Ana Miranda

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Reading in the data set:

```
cm <- read.table("./data/CH01PR20.txt", header = F)
colnames(cm) <- c("time", "copiers")
glimpse(cm)
```

```
## Rows: 45
## Columns: 2
## $ time    <int> 20, 60, 46, 41, 12, 137, 68, 89, 4, 32, 144, 156, 93, 36, 72, ~
## $ copiers <int> 2, 4, 3, 2, 1, 10, 5, 5, 1, 2, 9, 10, 6, 3, 4, 8, 7, 8, 10, 4, ~
```

```
summary(cm)
```

```
##      time      copiers
## Min.   : 3.00   Min.   : 1.000
## 1st Qu.: 36.00   1st Qu.: 2.000
## Median : 74.00   Median : 5.000
## Mean   : 76.27   Mean    : 5.111
## 3rd Qu.:111.00   3rd Qu.: 7.000
## Max.   :156.00   Max.    :10.000
```

Estimating the regression function:

```
cm_SLR <- lm(data = cm, time ~ copiers)
summary(cm_SLR)
```

```
##
## Call:
## lm(formula = time ~ copiers, data = cm)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.7723  -3.7371   0.3334   6.3334  15.4039
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5802     2.8039  -0.207   0.837
## copiers       15.0352     0.4831  31.123 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.914 on 43 degrees of freedom
## Multiple R-squared:  0.9575, Adjusted R-squared:  0.9565
## F-statistic: 968.7 on 1 and 43 DF,  p-value: < 2.2e-16
```

The estimated regression function is:

$$\widehat{time} = -0.5802 + 15.0352(copiers)$$

1 Problem 2.14 (a, b) (p. 91) (10 pts).

a) Obtain a 90 percent confidence interval for the mean service time on calls in which six copiers are serviced. Interpret your confidence interval.

```
predict(cm_SLR, newdata = data.frame(copiers = 6), se.fit = T, interval = "confidence", leve = 0.90)

## $fit
##      fit      lwr      upr
## 1 89.63133 87.28387 91.9788
##
## $se.fit
## [1] 1.396411
##
## $df
## [1] 43
##
## $residual.scale
## [1] 8.913508
```

From the output above we can see that the expected Y value for when the number of copiers is 6 is: $\hat{Y}_h = 89.6313$.

The standard error for the fitted value is: $s\{\hat{Y}_h\} = 1.3964$.

With $t_{critic}(0.95; 43) = 1.6811$ and solving the computation: $89.6313 \pm 1.6811 \cdot (1.3964)$. The 90% confidence interval for the expected Y value is:

$$87.28387 \leq E\{\hat{Y}_h\} \leq 91.9788.$$

Therefore, I'm 95% confident that the true mean value of Y, when $X = 6$, falls between 87.29 and 91.98 based on this regression model and data.

b) Obtain a 90 percent prediction interval for the service time on the next call in which six copiers are serviced. Is your prediction interval wider than the corresponding confidence interval in part (a)? Should it be?

```
predict(cm_SLR, newdata = data.frame(copiers = 6), se.fit = F, interval = "prediction", leve = 0.90)

##      fit      lwr      upr
## 1 89.63133 74.46433 104.7983
```

The prediction interval is: $74.46433 \leq E\{\hat{Y}_h\} \leq 104.7983$. Yes, it's wider than the corresponding C.I. in part a, as presented above, and that's how it should be. That is because the standard error calculated for the prediction interval includes one more term for the variance of the model than the usual C.I. does.

- 2 Review the example in Lab 1. R users see item 8 in RLab1_SLR.pdf. Practice drawing the “confidence interval band” and “prediction interval band” on the scatter plot with the regression line. Turn in a graph using Copier Maintenance data. (5 pts)

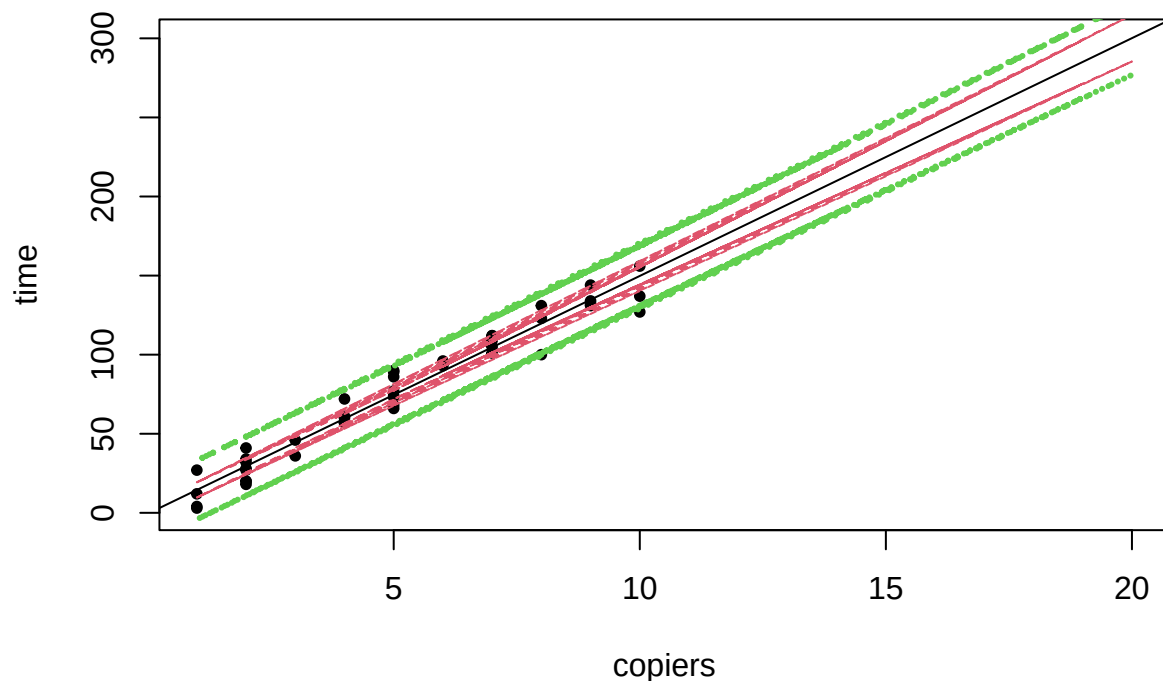
```
plot(time ~ copiers, data = cm, ylim=c(1, 300), xlim=c(1, 20), pch=20)
abline(cm_SLR$coef)
```

```
set.seed(1)
newx <- sample(1:20, size = 45, replace = TRUE)
newx
```

```
## [1] 4 7 1 2 11 14 18 19 1 10 14 10 7 9 15 5 9 14 5 5 2 10 12 15 1
## [26] 20 3 6 10 10 6 15 20 20 12 6 8 12 6 7 19 10 6 14 2
```

```
cm_CI <- predict(cm_SLR, newdata = data.frame(copiers = newx), interval="confidence", leve=0.95)
#cm_CI
cm_PI <- predict(cm_SLR, newdata = data.frame(copiers = newx), interval="prediction", leve=0.95)
#cm_PI
```

```
# lty=line types, col=colors for lines and points, lwd=line widths
lines(newx, cm_CI[, 2], lty= 2, col= 2) #Red for CI
lines(newx, cm_CI[, 3], lty= 2, col= 2) #Red for CI
lines(newx, cm_PI[, 2], lty= 3, col= 3, lwd= 3) #Green for PI
lines(newx, cm_PI[, 3], lty= 3, col= 3, lwd= 3) #Green for PI
```



3 Problem 2.24 (b, c, d) (p. 93) (15 pts).

a) Set up the basic ANOVA table in the format of Table 2.2. Which elements of your table are additive?

```
anova(cm_SLR)
```

```
## Analysis of Variance Table
##
## Response: time
##           Df Sum Sq Mean Sq F value    Pr(>F)
## copiers    1  76960   76960   968.66 < 2.2e-16 ***
## Residuals 43    3416     79
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The Regression Sum of Squares and the Residual Sum of Squares are additive to form the Total Sum of Squares measure. The Mean Squares values are computed dividing the Sum of Squares values by their respective degree of freedom.

b) Conduct an F test to determine whether or not there is a linear association between time spent and number of copiers serviced; use $\alpha = .10$. State the alternatives, decision rule, and conclusion.

Considering the F test to determine if there is a linear association between the variables `time` and `copiers`, the hypothesis test for the slope β_1 is the following:

- $H_0 : \beta_1 = 0$ vs $H_A : \beta_1 \neq 0$
- The F test-statistic calculated from the regression model is $F^* = MSR/MSE = 968.66$ and is provided by the ANOVA table above.

```
fcrit <- qf(p = 0.10, df1 = 1, df2 = (45-2), lower.tail = FALSE)
fcrit
```

```
## [1] 2.825999
```

With $\alpha = 0.10$ and $n = 45$, the F critic from the F distribution table is $F(0.90; 1; 43) = 2.826$. The decision rule is:

- If $F^* \geq 2.826$, conclude H_0
- If $F^* > 2.826$, conclude H_a

Therefore, H_A is concluded and $\beta_1 \neq 0$, meaning that there is a linear association between time spent in servicing copiers and number of copiers serviced. The p -value($Pr(> F^*)$) = $2.2e - 16$ and a lot smaller than 0.10.

c) By how much, relatively, is the total variation in number of minutes spent on a call reduced when the number of copiers serviced is introduced into the analysis? Is this a relatively small or large reduction? What is the name of this measure?

The total variation in number of minutes spent on call is relatively reduced by 95.75% when the number of copiers serviced is introduced to the model.

The name of the measure is **Coefficient of Determination: R-squared**. It can be calculated by $R^2 = SSR/SST = 0.9575$ or 95.75%. It's considered pretty large for this study, meaning that about 95.75% of the variation in service time can be explained by the number of copiers serviced.

d) Calculate r and attach the appropriate sign.

```
correlation = sqrt(0.9575)
correlation
```

```
## [1] 0.9785193
```

- The Coefficient of Correlation between the variables `time` and `copiers` is: $r = +0.9785$

4 Refer to the Copier Maintenance data. Learn to read basic output of regression analysis from another software. We will use SPSS output in the midterm exam.

- The SPSS output for regression analysis is similar to the output generated from R, just with a bit more information such as the SSTO on the ANOVA table and a few more labels as well.